# Noah Whittenbarger

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## Education and Training

**Meharry Medical College, School of Applied Computational Sciences**, Nashville, TN — Aug. 2025 - Present
Doctor of Philosophy — GPA: 4.00/4.00
Biomedical Data Science

**Meharry Medical College, School of Applied Computational Sciences**, Nashville, TN — Dec. 2025
Master of Science — GPA: 4.00/4.00
Biomedical Data Science

**NIH R25 Scholars Program,** Nashville, TN — May 2023 - Aug. 2023
Meharry Medical College
Mentor: Aize Cao, Ph.D.

**University of Cincinnati, McMicken College of Arts and Sciences**, Cincinnati, OH — Apr. 2017
Bachelor of Science — GPA: 3.52/4.00
Major: Biology - Biological Sciences
Minor: Chemistry

## Experience

**Meharry Medical College,** Nashville, TN — Mar. 2024 - Present
*Graduate Research Assistant – Dr. Saha Resonance Laboratory*
- Investigate the relationship between body mass index and human upper airway area using MRI based imaging.
- Use acoustic features and machine learning methods to predict human upper airway area.

**Vanderbilt University Medical Center,** Nashville, TN — Sep. 2020 – Jan. 2024
*Clinical Translational Research Coordinator II*
- Create 3D models of electrode array position using artificial intelligence for patients with cochlear implants.
- Conduct informed consent process, creation/completion of source documents and case report forms, administration of study drug, AE/SAE follow up, lab specimen processing and shipment, and communication with IRB, CRO, and study Sponsor.

**Vanderbilt University,** Nashville, TN — Jun. 2021 – May 2023
*Research Intern - Dr. Jamey Young Laboratory*
- Investigate metabolic processes that inhibit and promote fibrosis in mice with nonalcoholic fatty liver disease.
- Perform PCRs, western blots, and DNA and RNA extractions to accomplish tasks required by study design.

**Consultants for Clinical Research,** Cincinnati, OH — Jan. 2020 - Sep. 2020
*Clinical Research Coordinator*
- Independently manage six non-alcoholic steatohepatitis (NASH) studies and one primary biliary cholangitis (PBC) study.
- Conduct screening visits, informed consent process, creation/completion of source documents and case report forms, administration of study drug, AE/SAE follow up, lab specimen processing and shipment, and communication with IRB, CRO, and study Sponsor.

**Consultants for Clinical Research,** Cincinnati, OH — Aug. 2019 - Dec. 2019
*Recruitment Specialist*
- Screen patients for study eligibility based on study inclusion and exclusion list.
- Create and manage potential patient database.

**ScribeAmerica,** Cincinnati, OH — Sept. 2017 – Jun. 2018

*Medical Scribe - Ocular Oncology*

- Identify relevant information pertaining to physician's evaluations and document in medical note.
- Compile data from physical examinations and appropriately bill patients.

**University of Cincinnati,** Cincinnati, OH Oct. 2017 - Apr. 2018

*Supplemental Instruction Leader - General Physics I*

- Instruct sessions that reaffirm what students have learned in lecture.
- Hold weekly office hours to assist students in understanding concepts.

**The Christ Hospital,** Cincinnati, OH Jul. 2017 - Nov. 2017

*Patient Care Technician - Emergency Department*

- Undertook technical assignments such as drawing blood, conducting EKGs, and laboratory analysis.
- Assisted patients in activities of daily life and finding information.

**Good Samaritan Hospital,** Cincinnati, OH Mar. 2016 - Mar. 2017

*Patient Care Assistant - Geriatric/Psychiatric Unit*

- Executed various tasks including bladder scans, taking blood sugar, and measuring vital signs.
- Aided psychiatric patients with Activities of Daily Living (ADL) and perineal care.

**University of Cincinnati,** Cincinnati, OH Jun. 2015 - Jan. 2016

*Undergraduate Research Assistant*

- Genotyped putative transgenic lines using *X.laevis* models. Performed fluorescent microscopy, polymerase chain reactions, and polyacrylamide gel electrophoresis.
- Maintained *X. laevis* colonies, ordered supplies, and performed other lab maintenance as needed.

## Conference Abstracts

1. **Whittenbarger N**, Pinky L. Predicting Pharyngitis Etiology from Clinical Symptoms. Meharry Research Day; Nashville, TN; March 4-5, 2026.
2. Saha S, **Whittenbarger N**, Cao A. Effect of Obesity on Upper Airway Area During Vowel Articulation: An Mri-based Study. American Thoracic Society Conference; San Francisco, CA; May 16-21, 2025.
3. **Whittenbarger N**, White K, Kotun H, Saha S. Predicting MRI-Measured Upper Airway Area from Acoustic Features of Vowel Articulation. Bridge2AI Conference; Tampa, Florida; April 22-24, 2025.
4. **Whittenbarger N**, White K, Kotun H, Saha S. Predicting MRI-Measured Upper Airway Area from Acoustic Features of Vowel Articulation. Research Centers in Minority Institutions Conference; Bethesda, Maryland; March 16-19, 2025.
5. **Whittenbarger N**, White K, Kotun H, Saha S. Effect of Obesity on Upper Airway Area During Vowel Articulation. Meharry Research Day; Nashville, TN; March 5-6, 2025.

## Peer-Reviewed Publications

1. Smetak MR, Riojas KE, **Whittenbarger N,** Noble JH, Labadie RF. Dynamic Behavior and Insertional Forces of a Pre-curved Electrode Using the Pull-Back Technique in a Fresh Micro-Dissected Cochlea. (Accepted by Otology & Neurotology)

## Oral Presentations

1. **Whittenbarger N**, Cao A. Association of Maternal Health Outcomes with Cardiovascular Disease Using Survey Data. Oral presentation by Whittenbarger N at Fisk-Meharry-Vanderbilt NIH R25 Program Presentations; August 1-4, 2023; Nashville, TN.

## Podium Presentations

1. Smetak MR, Riojas KE, Noble JH, Labadie RF. Dynamic Behavior and Insertional Forces of a Pre-curved Electrode Using the Pull-Back Technique in a Fresh Micro-Dissected Cochlea. Podium presentation by Smetak MR at CI2022 American Cochlear Implant Alliance Meeting; May 18-21, 2022; Washington, DC.

2. Smetak MR, Riojas KE, Noble JH, Labadie RF. Dynamic Behavior and Insertional Forces of a Pre-curved Electrode Using the Pull-Back Technique in a Fresh Micro-Dissected Cochlea. Podium presentation by Smetak MR at 2022 AOS/COSM Spring Meeting Scientific Sessions; April 29-May 1, 2022; Dallas, TX.

## Technical Skills

**Laboratory Techniques:** Fluorescence microscopy, Polyacrylamide gel electrophoresis
**Programming Languages:** Python, C/C++, R, SAS
**Databases:** SQL, Tableau
**Operating Systems:** Linux, Mac, Microsoft Windows

## Extracurricular Activities

**MAPS: Minority Association of Pre-Medical Students**, University of Cincinnati — Aug. 2015 – Aug. 2016
*Community Service Chair*
- Organized community service events.
- Orchestrated over 150 hours of community service by members in the 2015-2016 school year.

**Delta Tau Delta Fraternity**, University of Cincinnati — Aug. 2015 – Aug. 2016
*Community Service Chair*
- Revised and coordinated community service events.
- Orchestrated over 500 hours of community service by members in the 2015-2016 school year.

**Caducea Pre-Medical Society**, University of Cincinnati — Aug. 2014 – Apr. 2017
*Active Member*
- Aims to expand importance of pre-medical education and scholarship.
- Promotes cooperation between students and faculty in different medical fields.